

20 Systems AUTOMATED PIPELINES End-to-end production tools	100+ Scripts PYTHON CODEBASE High-performance modules	600+ Files DOCUMENTS PROCESSED PDFs, CSVs, Excel manifests	100% Offline AI PRIVACY & SECURITY Air-gapped Ollama inference
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1. MULTI-CARRIER LOGISTICS & OCEAN FREIGHT

1. BL-ONE-SITC (Ocean B/L Draft & Invoice Reconciler)

Problem Solved & Contribution: Automates data extraction from Ocean Network Express (ONE) and SITC Draft Bills of Lading and multi-page attached commercial invoices (Tsusho). Solves multi-page address overflow (SH> / CN> continuation blocks), container/seal number pairing, and cargo weight/CBM cross-validation.

Tech Stack: Python, PyMuPDF (fitz), openpyxl (custom grid styling, freeze panes, wrap text), regex spatial coordinate clipping

2. CARGO NO (HS Code Extractor 1.0 – 1.2)

Problem Solved & Contribution: Extracts 8-to-10 digit Customs Harmonized System (HS) Codes and corresponding part numbers from complex Excel import instruction sheets (.xls / .xlsx).

Tech Stack: Python, pandas, regex pattern matchers, PyInstaller standalone executable

3. IMPORT CARGO (Import Cargo Cross-Referencing Tool)

Problem Solved & Contribution: Cross-references and merges data between Excel Import Instructions (.xlsx) and Chemiplas commercial invoice PDFs (.pdf), extracting internal cargo codes, tariff rates, and CSM approval numbers.

Tech Stack: Python, pdfplumber, pandas, win32com.client, openpyxl.utils, configparser

4. VN (TTAST Steel Export PDF Extractor 1.0 – 1.3)

Problem Solved & Contribution: Extracts steel coil export specifications, Japanese Industrial Standards (JIS), tax IDs, and dimensional measurements from TT Automotive Steel (Thailand) shipment PDFs into formatted Excel sheets.

Tech Stack: Python, pdfminer.six (LAParams, PDFParser), openpyxl, pathlib

5. VN-IPVL (Invoice & Packing List Dimension Extractor)

Problem Solved & Contribution: Parses steel manufacturing invoices to detect mill/slit edge specifications, industrial steel grades (SPFH590, TIS 1884-2564), and thickness/width dimensions into Excel.

Tech Stack: Python, pdfminer.six, openpyxl, regex dimension tokenizer

2. CUSTOMS, FINANCIAL & TAX EXTRACTION ENGINES

6. pdfToTax (Customs PDF to Tax Declaration System 1.0 – 1.10)

Problem Solved & Contribution: High-volume enterprise system processing 355+ Thai Customs declaration PDFs (QDRS...), extracting declaration numbers, CIF values, insurance, duty rates, and VAT breakdowns into audit-ready tax sheets.

Tech Stack: Python, pdfminer.six, openpyxl, regex data pipeline, PyInstaller

7. CHEMIPLAS INVOICE (Toyotsu Chemiplas Invoice & Packing Sheet Parser)

Problem Solved & Contribution: Batch converts multi-page commercial invoices and packing sheets into dual-sheet Excel workbooks with automated cell highlight alerts for missing data.

Tech Stack: Python, PyMuPDF (fitz), openpyxl (PatternFill), regex line tokenizers, AST parsers

8. Chemiplas_N (Composite PDF Invoice & Packing Splitter)

Problem Solved & Contribution: Intelligently splits composite master PDFs into separate Invoices and Packing Lists using coordinate-based visual boundary detection (`coord_finder.py`, `redline_xAxis.py`) and exports clean Excel reports.

Tech Stack: Python, `pdfminer.six`, PIL (Image coordinate visualization), PyMuPDF, `openpyxl`

9. CUSTOM INV (Customs Declaration & Tax Bill Parser)

Problem Solved & Contribution: Automates data extraction from Thai Customs Department declaration entries (ใบขนสินค้า / ใบเสร็จรับเงิน), Tax IDs, VAT values, customs fees, and payment dates into consolidated Excel records.

Tech Stack: Python, PyMuPDF (`fitz`), `pandas`, `openpyxl`, Thai Unicode text tokenizers

10. CSV INVOICE PACKING (CSV PackingList Aggregator 1.0 – 1.3)

Problem Solved & Contribution: Automates batch parsing and consolidation of over 100+ export packing list CSV files (EXPORTPAC_TCOM...) into standardized master Excel reports.

Tech Stack: Python, `pandas`, `xlsxwriter`, `zipfile`, PyInstaller

11. INVOICE EXTRACTOR (InvPackingList Extractor 1.0 – 4.1)

Problem Solved & Contribution: Multi-version commercial invoice extraction engine with automated multi-layout detection and error logging.

Tech Stack: Python, PyMuPDF (`fitz`), `pandas`, `openpyxl`, `tqdm` progress tracking, logging, PyInstaller

12. OT Extractor (Customs Overtime & Receipt Fee Extractor 1.0 – 1.2)

Problem Solved & Contribution: Scans Thai Customs electronic receipts (ใบเสร็จรับเงิน กรมศุลกากร), extracting officer overtime (OT) charges, clearance fees, and receipt numbers into summary cost reports.

Tech Stack: Python, `pdfminer.six`, `openpyxl`, `argparse`, logging

3. FREE TRADE AGREEMENTS (FTA) & REGULATORY COMPLIANCE

13. EPA JP (JTEPA Agreement Form JP Extractor 1.0 – 2.1)

Problem Solved & Contribution: Parses Japan-Thailand Economic Partnership Agreement (JTEPA) Form JP certificates (15+ pages), extracting exporter/importer details, product tables, and non-numerized part numbers into structured Excel files.

Tech Stack: Python, PyMuPDF (`fitz`), `openpyxl`, regex pattern engine

14. EPA-ATG KH (Cambodia E-Form D ATIGA Extractor)

Problem Solved & Contribution: Extracts tariff and shipment data from Cambodia-Thailand ASEAN Trade in Goods Agreement (ATIGA / E-Form D) certificates (e.g. NHK Spring Cambodia / Toyota Tsusho), mapping HS codes and invoice values.

Tech Stack: Python, PyMuPDF (`fitz`), `openpyxl` table styler

15. CO-JTCH (Certificate of Origin Item Extractor)

Problem Solved & Contribution: Extracts structured numbered item entries, steering parts, clamp/spring item codes, quantities, and HS codes from EPA Certificate of Origin PDFs.

Tech Stack: Python, PyMuPDF (`fitz`), `pandas`, `openpyxl`

4. APPLIED AI, VISION & OCR PIPELINES

16. Ollama_Model (Local LLM Document Intelligence)

Problem Solved & Contribution: Integrates local Large Language Models (`minstral3`, `ollama`) to perform zero-shot schema extraction, invoice header classification, and unstructured text normalization with 100% offline privacy.

Tech Stack: Python, `ollama` Python SDK, PyPDF2, PyMuPDF (`fitz`), `rich.console`, `ThreadPoolExecutor`, `requests`

17. OCR-Thing (Optical Character Recognition Engine)

Problem Solved & Contribution: Extracts text and tabular billing data from non-searchable, scanned paper invoices (e.g. Qingdao Special Iron & Steel) where direct digital text extraction is impossible.

Tech Stack: Python, EasyOCR, pdf2image, numpy, OpenCV image preprocessing

5. INDUSTRIAL SUPPLY CHAIN & SPECIALIZED MANUFACTURING

18. RETURNABLE RACK (Keyword PDF Filter & Splitter 1.0 – 1.3)

Problem Solved & Contribution: Bulk filters, keyword-searches, and splits/merges across 70+ delivery order (DO) PDFs, isolating pages for returnable container fleet logistics.

Tech Stack: Python, PyPDF2, PyMuPDF (fitz), glob, PyInstaller

19. JTEKT (JTEKT Automotive Parts Manifest Extractor)

Problem Solved & Contribution: Automates extraction and categorization of automotive steering parts, packaging units, and invoice line items based on custom product markers ('U' or 'K').

Tech Stack: Python, PyMuPDF (fitz), pdf2image, pandas, regex rule parser, PyInstaller

20. RUBBER (Chemical Certificate & Rubber Batch Extractor 1.0 – 1.1)

Problem Solved & Contribution: Extracts data from Thai chemical certificate forms (หนังสือรับรองยางผสมสารเคมี E-QC), parsing batch certificates (ECER...), compound formulas, test dates, and percentage compositions into Excel.

Tech Stack: Python, PyMuPDF (fitz), pandas, regex text slicing

PORTFOLIO METRICS & TECHNICAL SUMMARY

Total Automated Pipelines	20 Production Systems across 5 core industrial domains
Python Code Modules	100+ Custom Scripts and Production Automation Utilities
Enterprise Documents Handled	600+ PDF Declarations, CSV Packing Lists, and Excel Manifests
Accuracy & Reliability	Deterministic vector layout heuristics + Coordinate bounding-box clipping
Core Technologies	PyMuPDF, pdfminer, pandas, openpyxl, EasyOCR, Ollama LLMs, PyInstaller